

The relevance of tick bites to the production of IgE antibodies to the mammalian oligosaccharide galactose- α -1,3-galactose

Background In 2009, we reported a novel form of delayed anaphylaxis to red meat, which is related to serum IgE antibodies to the oligosaccharide galactose- α -1,3-galactose (alpha-gal). Most of these patients had tolerated meat for many years previously. The implication is that some exposure in adult life had stimulated the production of these IgE antibodies. **Objectives** To investigate possible causes of this IgE antibody response, focusing on evidence related to tick bites, which are common in the region where these reactions occur. **Methods** Serum assays were carried out using biotinylated proteins and extracts bound to a streptavidin ImmunoCAP. **Results** Prospective studies on IgE antibodies in three subjects following tick bites showed an increase in IgE to alpha-gal of twenty-fold or greater. Other evidence included i) a strong correlation between histories of tick bites and IgE to alpha-gal ($\chi^2=26.8$, $p<0.001$), ii) evidence that these IgE antibodies are common in areas where the tick *Amblyomma americanum* is common, and iii) a significant correlation between IgE antibodies to alpha-gal and IgE antibodies to proteins derived from *A. americanum* ($r_s=0.75$, $p<0.001$). **Conclusion** The results presented here provide evidence that tick bites are a cause, or possibly the only cause, of IgE specific for alpha-gal in this area of the United States. Both the number of subjects becoming sensitized and the titer of IgE antibodies to alpha-gal are striking. Here we report the first example of a response to an ectoparasite giving rise to an important form of food allergy.